SAMPLING-UNIT AND COMPOSITION SENSITIVITY OF AN AUDIO DEEPFAKE LEADERBOARD

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ABSTRACT

ASVspoof 2021 cites Bengio–Mariéthoz for its deepfake significance matrix without disclosing which test variant was used. Reconstructions separate five of six organizer-baseline pairs. With identical EER threshold refitting and all-pair multiplicity, trial-i.i.d. perturbation also gives 5/6, against 0/6 under speaker–attack product weights; a coherent marginal-sum covariance gives 0/6. An external fixed-family ASVspoof 5 check meets both registered descriptive criteria: at least one of six trial-i.i.d. exclusions disappears, and speaker–attack bands are typically more than twice as wide (observed median 27.27×). Separately, 12 internally frozen, score-blind composition policies reverse 6/12 within-block orderings but 0/16 contrasts between the four selected modern systems and four selected baselines. A post-failure constructive search, verified by three EER backends, finds reversal witnesses for all 12 within-block pairs, five at total variation $\leq$ .10 and the smallest at .010218. A frozen low-EER, zero-interaction stress cell has nominal-95% coverage .879/.880/.883, refuting universal calibration; 85.77% of spoof trials lie in blocks with only four or six speakers. We identify fixed-data evaluation-design sensitivity, not corrected population inference: large gaps between these selected blocks are stable over the registered policies, while their within-block orderings are not.

Index Terms— Anti-spoofing, audio deepfake detection, evaluation methodology, clustered bootstrap, ranking uncertainty

1. INTRODUCTION

In 2021 the ASVspoof organizers tested every pair of submitted systems for significance, applied a Holm–Bonferroni correction, and published the deepfake task's systems in four groups, noting that the top two were not separated [1]. We found no uncertainty analysis in three subsequent benchmark reports: the journal overview [2], ASVspoof 5 [3], and the recent multi-benchmark leaderboard [4] contain none of *significance test*, *confidence interval*, *bootstrap*, *p-value* or *standard error*. The sub-point margins on that modern leaderboard are therefore reported without an uncertainty test; a 2026 benchmark of self-supervised models likewise ranks point EERs without an interval, error bar or test [5]. The 2026 edition adds a calibration study instead; calibration establishes whether a score means what it says, not whether two systems differ.

That matters more than it would if the one analysis the field did run had represented the dependence. Both variants of the cited test [6] treat trials within a class as independent Bernoulli draws; the dependent form accounts for two systems sharing trials, not for trials sharing speakers and attacks. ASVspoof 2021 DF contains 533,928 evaluation trials but only 93 speakers and 110 attacks, and per-attack EERs for SSL-AASIST span 0.4–10.7%. Speaker ICC on bona-fide scores is 0.15–0.77, so repeated trials plainly do not supply independent information. ICC alone does not establish paired error correlation or an EER design effect; those are measured directly below by matched perturbations of the joint scores and then checked externally at family level on ASVspoof 5.

We vary two declared evaluation-design choices: the sampling unit used to perturb paired scores, and the observed source/task composition used to weight them. No statistical method is new—clustered resampling is standard [7, 8, 9]—and the contribution is the measured sensitivity and its audited boundary. The 5/6-to-0/6 change survives matched threshold re-estimation and a coherent covariance construction; explicit composition witnesses reverse within-block point orderings. A tail stress test refutes low-EER calibration, while corpus labels are associated with delete-speaker influences. The defensible result concerns outputs of specified procedures on released scores, not a replacement population verdict.

Scope. The primary pool comprises eight author or organizer releases with high provenance, not the universe of public re-scores. We analyze all eleven Speech DF Arena re-scores [4] separately. No full-21DF "resolved" label transfers to new speakers or attacks: four source/task blocks lack speaker replication. Bands and composition radii are descriptive procedure outputs on fixed scores, not population confidence or safety claims.

2. RELATED WORK

Challenge-ranking stability. Ranking sensitivity to test cases, metrics, aggregation and reference annotations is established in biomedical challenges [10]; challengeR uses bootstrap views to expose data and ranking variability [11]. Our narrower contribution is an audio-deepfake application that aligns EER threshold refitting and familywise comparisons, audits crossed speaker–attack perturbations, and constructs source/task composition witnesses.

This benchmark's own reporting. ASVspoof 2021 pairs systems with the test of Bengio and Mariéthoz [6] under Holm correction and publishes group structure [1]; Wang and Yamagishi [12] showed that changing only the training seed moves ADD rankings—a variance component our intervals do *not* include.

Clustered resampling. That the *number of users*, not of samples, governs a biometric error rate's uncertainty is due to Poh, Martin and Bengio [7], who bootstrapped over users and samples and extrapolated a DET curve to eight times more users—and set our case aside explicitly: "this issue is not the utmost concern when comparing two systems evaluated on the same sets of users… It becomes a concern when different users are involved. The latter subject is our focus." Wu et al. carried it into speech, showing on NIST SRE that enlarging the sample cannot reduce dependence's effect on a detection cost's standard error [13], and building the paired correlation-corrected two-system test [8]—whose clustered analysis created a zero-exclusion where uncorrected intervals overlapped, whereas our declared perturbation removes zero-exclusions. Both name EER as an unexecuted extension. Bolle et al. [14] proposed the subsets bootstrap for dependent biometric errors; Fogliato et al. [15] found that family under-covers for 1:1 matching, where dependence is *dyadic*—each score compares two images, so resampling distorts the comparison graph. Each ASVspoof trial is indexed by one speaker and, for spoof trials, one attack, yielding two crossed clustering factors [16], neither nested nor dyadic; crossed designs are the delicate case [17, 18], and a hierarchical bootstrap can beat the plain cluster form on nested metrics [19]. Bootstrap intervals for speech evaluation go back to Bisani and Ney [20], with blockwise forms for ASR [9]. What is new is the audited fixed-data sensitivity: after threshold refitting and all-pair multiplicity are aligned, zero-exclusion outputs change with the declared perturbation unit. We neither attribute that contrast to individual cells of the published greyscale figure nor treat the resulting bands as population confidence claims.

Audio and benchmark audits. Zhuang et al. [21] compute a paired-binomial detectable effect for quantization benchmarks, treating items as independent—the assumption under test here. In audio deepfakes, bona-fide cross testing shows aggregation weights change conclusions [22]; other audits question representations [23] and corpus overlap [24]. These works do not combine a named crossed dependence unit with pairwise composition-reversal witnesses. Simultaneous rank intervals are established [25]; adjacency is score-selected, so we control multiplicity over the whole family.

3. METHOD

3.1. Identified object: fixed-data procedure sensitivity

Two deterministic systems on a frozen trial list have *no* sampling uncertainty; an interval exists only after declaring what was sampled. The organizers' test treats trials within class as independent draws [6]. Our original working target instead treated speakers and attacks as exchangeable with a represented evaluation design. That law is not defensible here: source/task stratification leaves almost no empty cells, yet the VCC blocks contain only 4/4/4/6 spoof speakers, and corpus labels are associated with delete-speaker influences. We therefore do not replace the organizers' population claim with another unidentified one. The identified object is how paired ΔEER outputs on the fixed score set change under declared trial-i.i.d. and speaker–attack perturbation laws; confidence language is reserved for a future target with a defensible sampling law and replication.

3.2. Clustered bootstrap and simulation-checked variants

Speakers and attacks are resampled independently with replacement and trial weights multiply the two counts; bona fide trials carry no attack label and so receive speaker weights only. The primary output is the two-way percentile bootstrap. Exact delete-one refits separately compute $\Sigma_s + \Sigma_a - \Sigma_{sa}$ over 93 speakers, 110 spoof attacks and 1,062 observed spoof speaker$\times$attack cells [16]. Its former pairwise floor $\max(V_s + V_a - V_{sa}, V_s, V_a)$ did not define a joint covariance and is retired from simultaneous labels. The replacement corroborator uses the single PSD matrix $\Sigma_+ = \Sigma_s + \Sigma_a$: it dominates both marginals and the raw exact-cell matrix in Loewner order, and the same Σ_+ supplies every pair SE and the Gaussian max-t critical value ($q = 2.878$). It deliberately retains the cell overlap and is a coherent marginal-sum sensitivity covariance, not an exact multiway estimator or coverage claim. Product and coherent outputs agree on all 28 zero-exclusion labels (18/28; organizer 0/6). A smooth influence-function CRVE failed its internally frozen finite-refit gates on every pair and is excluded.

3.3. All-pair multiplicity for procedure bands

From B=5000 clustered replicates we form sup-t simultaneous procedure bands [26]: the critical value is the 95th percentile of $\max_p |\Delta_p^{(b)} - \bar{\Delta}_p|/s_p$ over replicates, applied to every pair. Rank sets are derived mechanically from those bands. Because the required sampling law is not identified for full 21DF, neither is called a population confidence set. A bootstrap rank percentile is separately released as a stability interval.

Reconstructing the cited test. The overview does not disclose which Bengio–Mariéthoz variant produced its greyscale matrix, so we make no cell-level attribution. With each system's non-interpolated EER threshold selected on the evaluation set, the independent and paired-decision reconstructions separate the same five of six baseline pairs. Holm is step-down, but the result is family-size independent: in the independent reconstruction the weakest significant pair has $p = 9.7 \times 10^{-6} < 0.05/561$, while the sixth has $p = 0.217 > 0.05$. Because the cited derivation assumes thresholds fixed outside the test set, these are reconstructions of its adaptation to EER rather than exact finite-sample tests.

3.4. Coverage check

A bivariate random-effects model generates scores *on the real trial index*—the actual speaker and attack of every trial, preserving the sparse unbalanced incidence but imposing an exchangeability law that is not defensible for the observed source/task structure. The internally frozen 48-cell grid crosses organizer-like/low EER, interaction shares 0–.75, Gaussian/Student-$t(5)$ effects and $\Delta \in \{0, .5, 2\}$, with high-precision population truth. The three then-candidate variants cover within $[.92, .98]$ in all 24 organizer-like model cells (ranges .928–.957 exact-cell and .946–.968 product). In low EER their minima are .845/.845/.855, and all three fall below .90 together in 11 of 24 cells. The frozen reading rule is therefore Refuted. An oracle constant-SD diagnostic stays at or above .944, while replicate-specific SE shrinks in extreme negative-tail samples; the defect is asymmetric non-pivotality, not average variance scale. Passing organizer-like cells validates implementation behavior under that DGP, not the DGP as a model of 21DF.

3.5. Composition sensitivity on fixed scores

Three bona-fide policies (empirical, source-balanced, source–speaker-balanced) and four spoof policies (empirical through attack-family-cell balanced) form a frozen 12-policy factorial. All trial weights remain positive and class-normalized. For each pair we report the policy range and registered empirical-to-policy paths; r_{TV} is the larger class-conditional total-variation shift. These are deterministic estimands, not confidence intervals. After two failed numerical implementations, a separately frozen post-failure search may only exhibit constructive upper bounds: a witness must reverse under count-canonical, direct-trial and tie-safe EER evaluators. Search failure is not stability; a reported bound is not a global minimum.

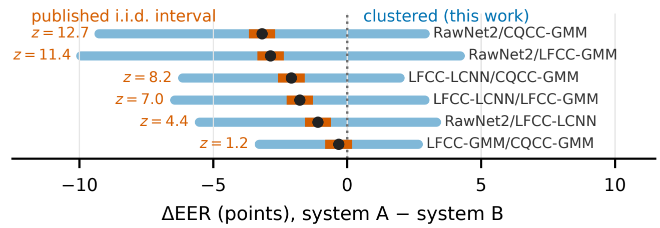

Fig. 1. The six baseline pairs, which appear in both the published significance figure and our pool. Narrow bars: the independent Bengio–Mariéthoz reconstruction (its z at left); paired-decision reconstruction gives the same five verdicts. Wide bars: the simultaneous speaker–attack procedure output. Dots: empirical gaps; all six wide bars straddle zero.

4. EXPERIMENTS

Setup. The primary pool contains eight high-provenance ASVspoof 2021 DF releases: four self-supervised systems (XLSR-Mamba [27], XLS-R+SLS [28], XLSR-Conformer [29], SSL-AASIST [30]) from their authors and four organizer baselines (RawNet2, LFCC-LCNN, LFCC-GMM, CQCC-GMM) from the challenge key package. We use the 533,928 eval-phase trials of 611,829 released. Pooled EERs run 1.88–2.85 for the self-supervised systems and 22.38–25.56 for the baselines, reproducing published values where they exist. Pairwise intervals use B=5000 clustered replicates (seed 20260817); the 48-cell coverage stress test uses B=500 per replicate over R=1000 replicates per cell (seed 20260824).

Sampling-unit sensitivity (Fig. 1). We reconstruct both variants cited by the overview from the released scores and claim no cell-level agreement with its greyscale figure. Each reconstruction separates the same five of six organizer pairs. More directly, a post-audit matched bootstrap uses the identical non-interpolated weighted EER, re-estimates its threshold in every replicate, and applies the same all-28-pair max-t rule in both arms: trial-i.i.d. perturbation gives 26/28 and 5/6 organizer zero-exclusions, against 18/28 and 0/6 under speaker–attack product weights. Thus threshold treatment does not explain the change. The coherent marginal-sum output separately gives 18/28 and 0/6. For the widest 3.18-point gap, the product band is $[-9.27, 2.91]$ and the coherent band $[-8.76, 2.40]$.

What the benchmark supports (Table 1). All six organizer pairs straddle zero under product and coherent marginal-sum perturbations despite a 3.18-point span. RawNet2–CQCC-GMM, LFCC-LCNN–LFCC-GMM and LFCC-LCNN–CQCC-GMM become zero-excluding in leave-one-corpus-out refits, so the organizer result is not composition-robust. The 16 gaps between our selected modern and baseline blocks are 19.5–23.7 points and survive every deletion. Two modern zero-exclusions at 0.97 and

Table 1. Pairs within the selected modern and baseline blocks: empirical gap and product-perturbation half-width (EER points), plus the smallest verified constructive r_{TV} upper bound. Printed bounds are rounded approximations to found reversals, not minima or confidence limits.

pair		gap	half-w.	r_{TV} bound
XLSR-Mamba	XLS-R+SLS	0.032	0.452	0.010
XLSR-Mamba	XLSR-Conf	0.389	0.619	0.343
XLSR-Mamba	SSL-AAS	0.968	0.555	0.245
XLS-R+SLS	XLSR-Conf	0.357	0.752	0.172
XLS-R+SLS	SSL-AAS	0.935	0.561	0.434
XLSR-Conf	SSL-AAS	0.578	0.674	0.165
RawNet2	LFCC-LCNN	1.094	2.868	0.069
RawNet2	LFCC-GMM	2.864	4.638	0.093
RawNet2	CQCC-GMM	3.180	3.952	0.111
LFCC-LCNN	LFCC-GMM	1.770	3.074	0.087
LFCC-LCNN	CQCC-GMM	2.086	2.636	0.104
LFCC-GMM	CQCC-GMM	0.316	1.996	0.033

0.94 points are withdrawn because low-EER stress is anti-conservative and both disappear after source deletion; adjacent modern margins are only 0.03, 0.36 and 0.58 points.

Composition robustness. Under the 12 registered score-blind policies, 6/12 within-block orderings reverse and 0/16 selected cross-block contrasts do; only one registered path reverses at $r_{TV} \leq .10$, yielding the frozen classification *Local composition fragility*. The later constructive search cannot revise that classification: it verifies 729 directions and bounds all 12 within-block reversals from above, five at $r_{TV} \leq .10$. XLSR-Mamba versus XLS-R+SLS reverses at .010218 with odds 1.073 and a .000699-point margin. Four other modern witnesses need odds above 3,000, so these are boundary-existence results, not practical ranking instability. No cross-block witness was found, which is a search outcome rather than proof of global stability.

Independent re-score layer. All eleven complete Speech DF Arena files map one-to-one to the same 533,928 trials, but remain a separate stratum because they are re-scores and same-named implementations need not match: Arena RawNet2 has 40.67% EER against 22.38% in the organizer release and score correlation 0.36. Applying the same procedure descriptively to the full 55-pair family changes the count of zero-excluding bands from 52 under trial-i.i.d. inference to 38 after speaker$\times$attack resampling; two score-adjacent edges change labels, and the median clustered/i.i.d. pointwise-width ratio is 8.45. These are provenance and sensitivity measurements, not coverage-validated Arena confidence claims.

Corpus-associated speaker influence. The bona-fide pool spans three source corpora (67, 12 and 14 speakers); grouping delete-speaker influences by corpus puts 9–48% of their finite-A sum of squares between groups (median 26%). A permutation test rejects corpus-label permutability for all twelve within-block pairs after Holm correction ($p_{\mathrm{Holm}} \leq 0.041$). This establishes provenance structure in this finite component, not failure of every exchangeability law or a speaker-count asymptote. Leave-one-corpus-out refits preserve all 16 selected cross-block verdicts; both within-SSL resolutions fall and three baseline pairs resolve in at least one refit. Thus every within-block resolution is provenance-conditional; the large selected-block separation is stable only under these deletions.

External family-level check. ASVspoof 5 replaces about a hundred studio-recorded speakers with 737 crowdsourced ones and cuts attacks from 110 to 16 [3]. We froze SSL-AASIST, AASIST, XLS-R+SLS and XLSR-Mamba (EER 16.25, 35.53, 18.76 and 14.40%) over all 680,774 trials (367 target and 370 bona-fide-only speakers). Criterion A required at least one trial-i.i.d. exclusion to disappear; B required median width inflation ≥ 2. Both hold. Role-stratified speaker–attack perturbation changes SSL-AASIST versus XLS-R+SLS from $[-2.657, -2.378]$ to $[-6.274, 1.238]$ EER points. Across the fixed six-pair family, bands are 11.23–35.90$\times$ wider (median 27.27$\times$); the focal ratio is 26.92$\times$. This is not pair-level replication: on 21DF that pair excludes zero under both arms, and the 5/6-to-0/6 organizer family has no ASV5 analogue. The acquisition-law gate is NO-GO; these are fixed-roster procedure outputs, not population confidence or significance.

Measured width inflation. On the real score files, simultaneous product-perturbation bands are 4.13–11.36$\times$ wider than their matched trial-i.i.d. counterparts over all 28 pairs (median 8.14$\times$). This requires no ICC approximation: both arms recompute the same EER threshold from shared system weights, and their saved clustered labels reproduce the primary analysis exactly. In the fitted organizer-like DGP on the real incidence, the nominal-95% i.i.d. interval covers its model population ΔEER 16% of the time against 95.5% and 96.5% for the then-candidate variants. That calibrates the imposed DGP, not 21DF's unidentified superpopulation.

5. DISCUSSION

For reporting, and limits. We release procedures, plans, deviations and an audit package covering all 28 21DF comparisons and the fixed-family ASVspoof 5 check—score-file hashes, available receipts, intersections, EER rule, weights and seeds—at `github.com/rvirgilli/asvspoof2021-df-clustered-uncertainty`. The legacy NPZ snapshots have file hashes but not historical run provenance, and the internal freeze has no independently verifiable public timestamp. Report speaker/attack incidence and composition mass, not trial counts alone. A band needs a sampling law; a composition radius needs an explicit policy family and witness. We therefore withdraw full-21DF population labels, modern rank confidence and formal Arena bands. The conclusion is fixed-data sensitivity: 5/6 versus 0/6 under two perturbation laws, not corrected population inference.

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